Intro to Computer Programming
Final Exam

Date	27 October 2026
Duration	1 week
Class	CPro2
Total Score	100
Test Format	Individual take home project

Overview

The final exam is a take-home project with two parts. **In Part 1 you build a menu-driven Python program that explores a real dataset from Kaggle**. **In Part 2 you record a video that demonstrates the program and explains which programming concepts you applied**.

The exam checks that you can use the topics of the course together in one working program, and that you understand what you built.

Submission

Submit one zip file that contains one folder with these items.

- All of your .py files and the CSV file in one zipped folder
- A pdf file report that contains: the dataset name, Kaggle link, why you chose it, the columns you use and skip, how to run the program and the tests, what each feature does, known limitations, the link to your explanation video, and a table that lists, for each concept, one file and function where you used it.

Your instructor must be able to run python main.py and python -m unittest / pytest from inside your folder with only Python 3 installed. *Leave out __pycache__ folders*.

Part 1: Project application requirements

Build a menu-driven program in Python 3 that explores a real dataset from Kaggle. You choose the dataset and the features. The subsections below give the details.

Dataset

Pick a dataset on Kaggle (kaggle.com/datasets) that meets all of these rules:

- **Shape:** one CSV file where each row is one item (a player, movie, book, car, song, city and so on), with 200 to 2,000 rows (If it exceeding 2000 row, you can trim it to fit the requirement).
- **Source:** real data whose license allows educational use, with no personal information about private individuals.
- **Blank cells:** check which columns are filled in for every row. You may use only the complete columns. If you use a column with blanks, explain in your README how the program treats them.

Program features

The program runs with python main.py, loads the CSV, prints how many records it loaded, and shows a menu that repeats until the user chooses Exit. What the menu offers is up to you. Include at least 3 different features that fit your data, for example search, listing a category, top N, comparing two records, summarising a category, a watchlist, filters or recommendations. Each feature must work correctly on your data.
Whatever you build, the program must follow these rules:

- Every typed value is checked. Bad input gives a clear message and a new prompt, never a crash.
- Rows that cannot be read are skipped and counted, and the program reports how many.
- Results appear in aligned columns, not as raw Python lists.
- Nothing divides by zero, and any search ignores capital letters and extra spaces.

Python concepts

Your code must use each concept below in a meaningful way, not in one throwaway example. The table lists the minimum for each concept.

Concept	**Example of your project must show**
Variables and data types	Use str, int, float, bool or list. Keep fixed values, such as the CSV file name, in constants written in capital letters.
Input and output	Read input with input(), validate it in a loop, and print results with f-strings, including aligned columns.
Operators	Use arithmetic operators to calculate at least two derived values (for example a rate or an average), relational operators to compare values, and and, or, not to combine conditions.
Conditional statements	Use if, elif and else to choose between at least three outcomes, for example when

	handling bad input or placing a record in a category.
Loop statements	Use for loops to process records and a while loop for the menu.
Functions	Write functions with parameters and/or return values.
Lists	Store all records in a list, build new lists for results such as filtered or ranked records, and use slicing (for example the first N items).
String manipulation	Write a function that cleans text, e.g. comparison (lower case, strip spaces, remove accents or hyphens) and use string methods such as split, replace, strip and lower.
Classes (OOP), optional	Bonus: write at least one class with __init__ and __str__ and use it to store each record. A second class that holds a list of records earns extra credit.
Unit tests	Write at least 5 tests with the unittest module. Cover normal input, edge cases (such as zero or an empty list) and bad input.

Use only the Python standard library, such as csv, os and unittest. Do not use pandas or numpy. If your program ranks or sorts records, write that loop yourself instead of calling sorted(), so that the loop is yours to explain.

Code structure and tests

Use at least four .py files, split by job. An inspiration structure that you can use:

- models.py (record class and constants),
- text_utils.py (string helpers),
- data_loader.py (reading the CSV),
- analysis.py (the functions behind your features),
- main.py (menu, input and output)
- test_project.py (unit tests).

Put the CSV in the same folder as main.py.

Part 2: Video explanation

You also submit a video in which you demonstrate your program and explain which programming concepts you applied and where. It shows that you understand what you built.

What the video shows

- **Length:** no longer than 15 minutes. Only the first 15 minutes are graded.
- **Demo:** start with a short demonstration of the running program. Show each feature once, including what happens with invalid input.
- **Concepts:** explain what programming concept that you applied to your built program with the code snippet location
- **Format:** a screen recording with your voice. Showing your face is optional. Reading a script aloud, or reading code line by line without explaining it, earns few marks.

Upload and sharing

Upload the video to YouTube, set its visibility to Unlisted or Public, and put the link in your README or pdf report. Unlisted means only people with the link can watch, and the video does not appear in search or on your channel. Do not choose Private, because your instructor cannot open a Private video.
Do not delete the video or change its visibility until your grade is released. If you cannot use YouTube, contact your instructor before the deadline to agree on another way to share it.

Grading Criteria

The project is graded out of 100 points. Bonus points for classes are added to your total, and the final mark is capped at 100.

Part	**Points**	**Full marks**
Dataset and loading	8	The dataset meets all the rules and is approved. The loader reads the file, skips bad rows and reports how many were skipped.
Program features	24	At least 3 different features work correctly on your data, input is validated and output is aligned and readable.
Use of the core concepts	24	Each of the eight required concepts (variables, input and output, operators, conditionals, loops, functions, lists, strings) is used meaningfully. Each is worth 3 points.

Unit tests	6	At least 5 tests, all passing, covering normal, edge and bad input.
README	4	Complete and accurate, and the run instructions work as written.
Video explanation	34	A working demo (9 points), each required concept located in your code and explained correctly in your own words (16 points), and a clear video within the time limit with readable code on screen (4 points). 5 points for engagement of your video: at least get 5 likes and 5 views
Bonus: classes (OOP)	Up to 10	A record class with __init__ and __str__ (5 points) and a second class that holds a list of records (5 points).

FAQ

- ***I am in stealth mode and do not want the people around me to know that I am pursuing a bachelor's degree. Can I submit my video in a different way?***
  You can set your video to Unlisted. An unlisted video does not appear in search results or on your channel, and only people who have the link can watch it. We strongly encourage you to publish your video publicly, so that you can share what you have learned with others and build your online presence, which may open the door to interesting opportunities. If you really cannot upload your video to YouTube, contact your lecturer directly to discuss an alternative.

- ***Can I publish my video on a social media platform other than YouTube?***
  Yes, as long as the platform supports videos of up to 15 minutes.

- ***Should I mention my class or university name in the video?***
  No, we encourage you not to. Make the video as natural as possible, so that viewers feel you truly want to share what you have built and what you know, rather than making a video only for the sake of the exam.

- ***I decided not to make my video public. Does that mean I cannot get the 5 engagement points?***
  Not necessarily. You can still share your unlisted video with friends to get at least 5 views and 5 likes.

- ***Can I use AI to complete this exam?***
  Yes, you may use AI to help write your code, but you must understand the code it generates. That is why this exam requires an explanation video, it lets us check that you really understand the code you submit.

- ***Can I submit the source code via GITHUB public repository and the report through readme.md instead of submitting an zip?***
  yes, please submit the GIT public link for it

Useful Resources:

- How to set video unlisted on youtube: https://www.youtube.com/watch?v=JOr7JluzEOM
- How to setup Github: https://www.youtube.com/watch?v=mJ-qvsxPHpY
- How to share Public Github repository: https://www.youtube.com/watch?v=UhttT2D42O4
- How to use Kaggle Dataset: https://www.youtube.com/watch?v=krkS9u140tM